PCIe x4 a 4x NVMe M.2

Ver: 1.0

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Power

LMZ31530 LMZ31704
AP7361C-ADJ

Packet Switch

PI7C9X3G808GP

Reset (PERST#)

74HC08

Bidirectional Level Shifter

PI4ULS5V108LEX

M.2 M-key 2280 Socket

J44 J45 J46 J47

PCIe x4 Gold Fingers

P2 (upstream)

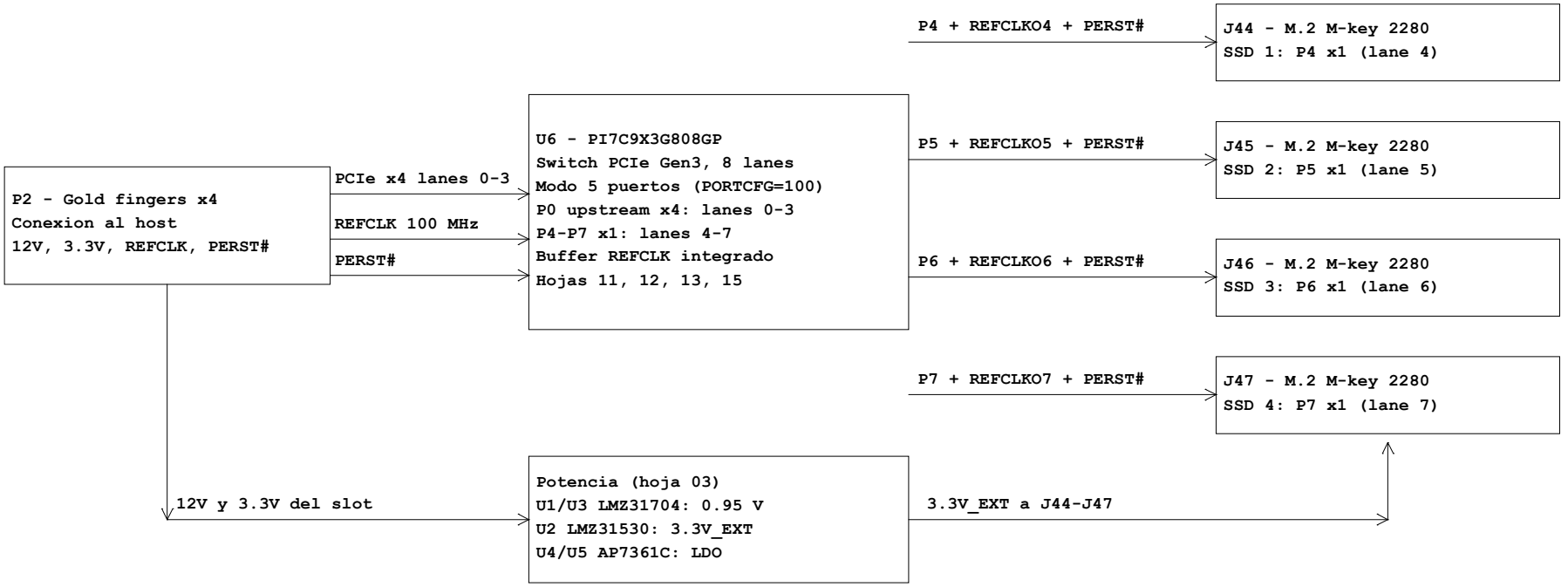
Removidos del EVB

U12-U17, S1-S3 ATX, U7

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Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size: Custom	Document Number: 01_COVER SHEET	Rev: 1.6
Date: Sunday, November 08, 2026	Sheet 1 of 20	

FUNCTION BLOCK - PCIe Gen3 x4 a 4x SSD NVMe M.2 (Rev 1.0)



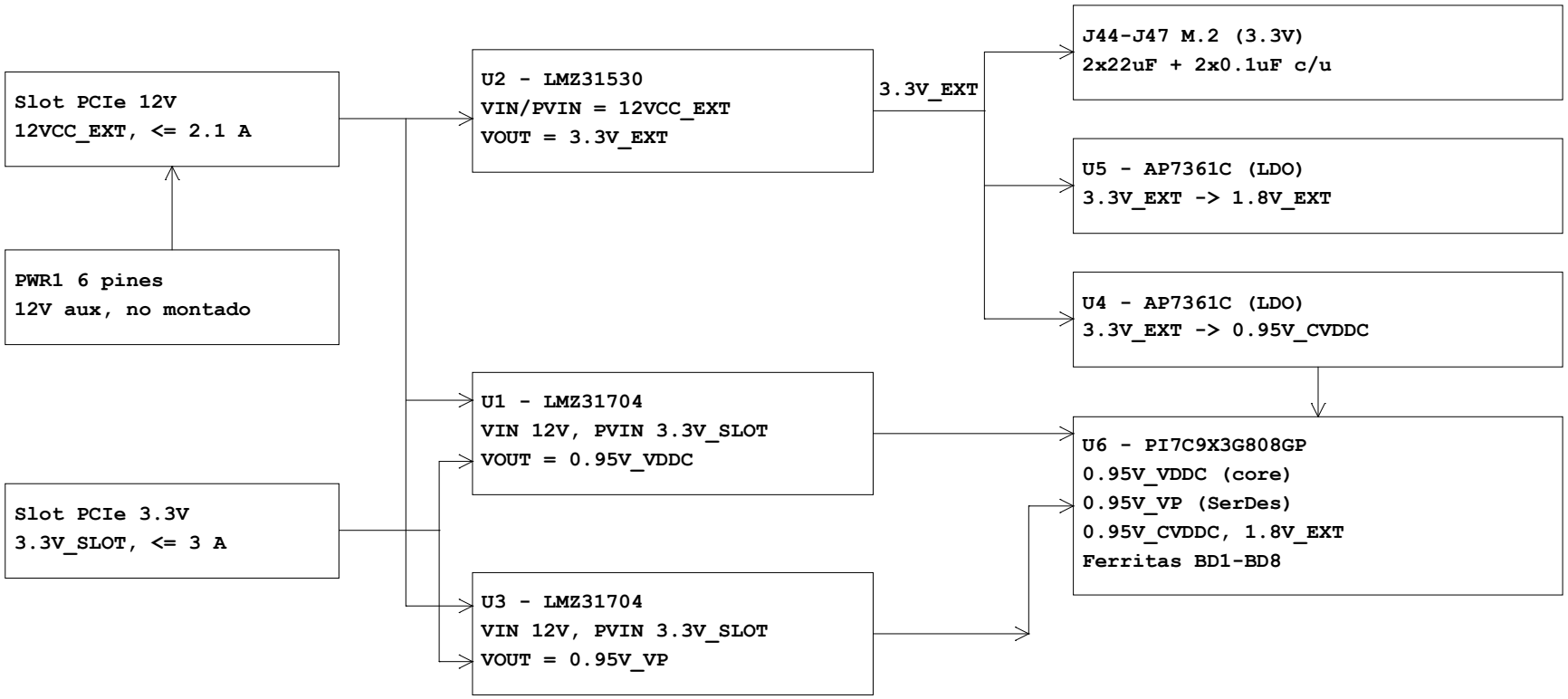
Cambios respecto al EVB Diodes V1.6: sin slots S1-S3, sin signal switches U12-U17, sin fuente ATX.

Conexion directa: gold fingers x4 -> U6; L4->J44, L5->J45, L6->J46, L7->J47.

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Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size C	Document Number 02_FUNCTION BLOCK	Rev 1.6
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POWER DELIVERY - arbol de potencia (Rev 1.0)

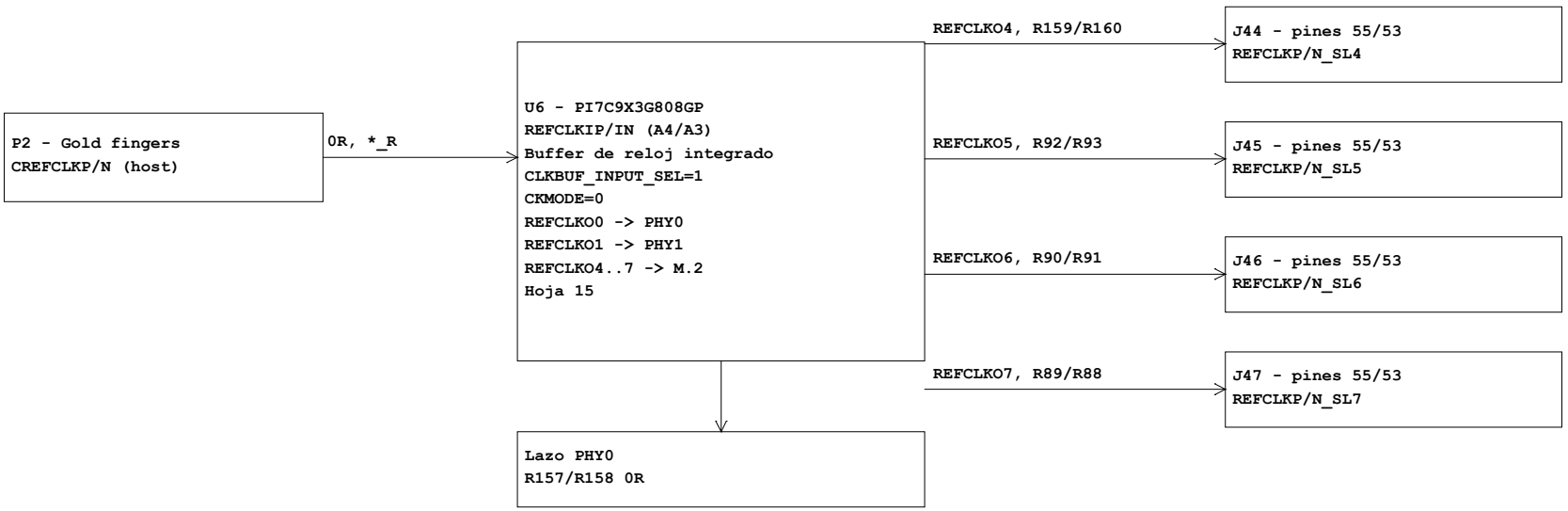


Presupuesto slot x4: 25 W (3.3V <= 3 A, 12V <= 2.1 A); 4 W por SSD.
0.95V_VDDC y 0.95V_VP en planos (TOP/ART04) en el layout.

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Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
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REFCLK BLOCK - reloj de referencia 100 MHz (Rev 1.0)



Sin generador ni buffer externo (U20 del EVB eliminado).

Pares de 85 ohm diferencial; igualacion intra-par <= 5 mil.

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Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size: Custom	Document Number: 04_REFCLK BLOCK	Rev: 1.8
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Port-Lane Mapping

A. Set Functional Mode

PORTCFG[2:0]

PORTCFG[2]	PORTCFG[1]	PORTCFG[0]	Functional Mode
0	0	1	2Port - 8Lane Configuration
0	1	0	3Port - 8Lane Configuration
0	1	1	4Port - 8Lane Configuration
1	0	0	5Port - 8Lane Configuration
1	0	1	8Port - 16Lane Configuration

Chipmode[1:0]

Chipmode1	Chipmode0	Mode
0	0	Normal
0	1	Reserved
1	0	Reserved
1	1	Reserved

C. Ports - Lanes mapping for different mode

MODE	2Port	3Port	4Port	5Port	8Port
Lane 0	P0	P0	P0	P0	P0
Lane 1	P0	P0	P0	P0	P1
Lane 2	P0	P0	P1	P0	P2
Lane 3	P0	P0	P1	P0	P3
Lane 4	P4	P4	P4	P4	P4
Lane 5	P4	P4	P4	P5	P5
Lane 6	P4	P5	P5	P6	P6
Lane 7	P4	P5	P5	P7	P7

B. Este diseno: PORTCFG=100 (5 puertos), CHIPMODE=00

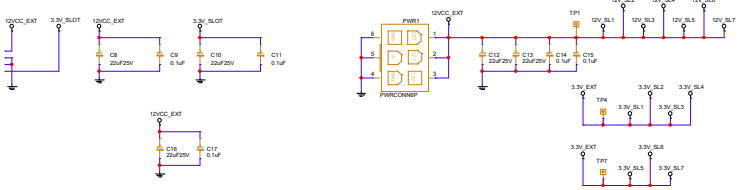
Lanes	Puerto	Conector	TX (C AC -> pin)	RX	REFCLK	PERST#
0-3	P0 x4 up	P2 fingers	PET0-3_P0	PER0-3	CREFCLK	CPERST_L
4	P4 x1	J44	C741/2 -> 47/49	43/41	55/53	50
5	P5 x1	J45	C751/2 -> 47/49	43/41	55/53	50
6	P6 x1	J46	C761/2 -> 47/49	43/41	55/53	50
7	P7 x1	J47	C771/2 -> 47/49	43/41	55/53	50

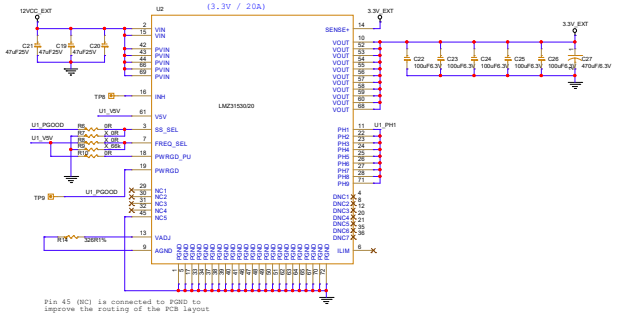
Sin signal switches: conexion directa.

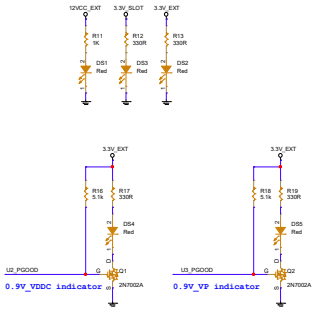
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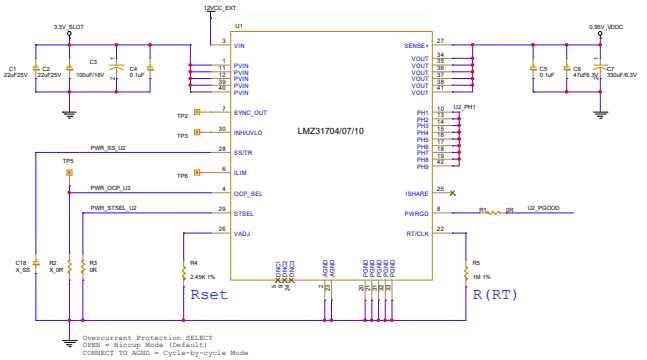
Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size: Custom	Document Number: 05_PORT-LANE MAPPING	Rev: 1.6
Date: Sunday, November 08, 2026	Sheet 5 of 20	

ATX POWER CONNECTOR







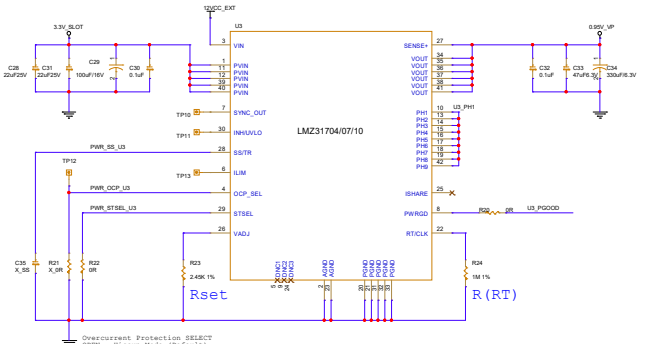


Slow-Start Capacitor Valueand Slow-Start Time							
Css (nF)	OPEN	3.3	4.7	10	15	22	33
SS Time (msec)	1.2	2.1	2.5	3.8	5.1	7.0	9.8

Power module Setting: (Vin=5V)

Vout = [(1.43 / Rset) + 1] x 0.6

Vout	Rset	R(RT)	fsw
0.95V	2.45K	1M	250KHz



Slow-Start Capacitor Valueand Slow-Start Time							
Css (nF)	OPEN	3.3	4.7	10	15	22	33
SS Time (msec)	1.2	2.1	2.5	3.8	5.1	7.0	9.8

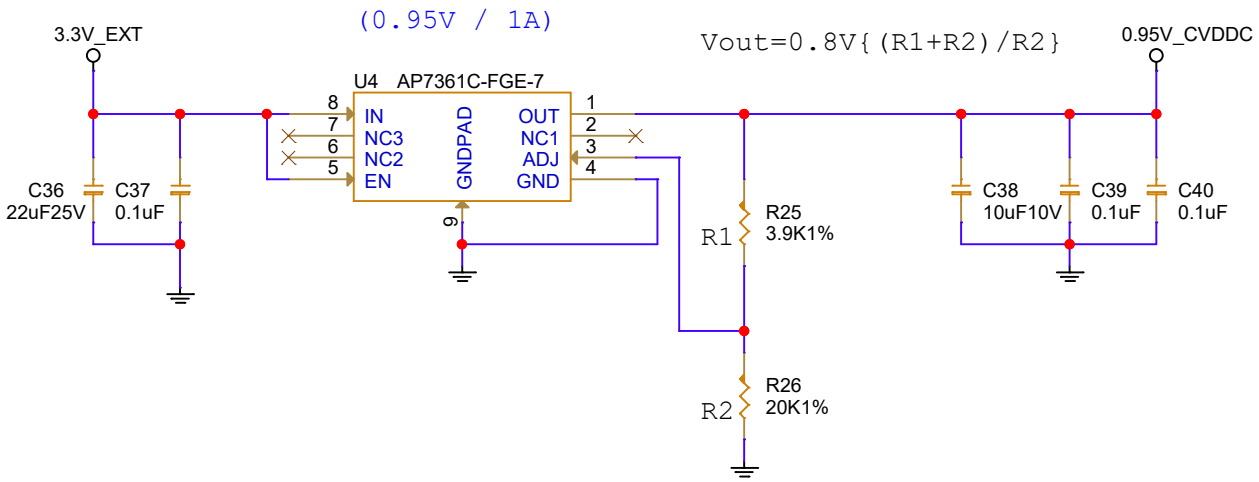
Power module Setting: (Vin=5V)

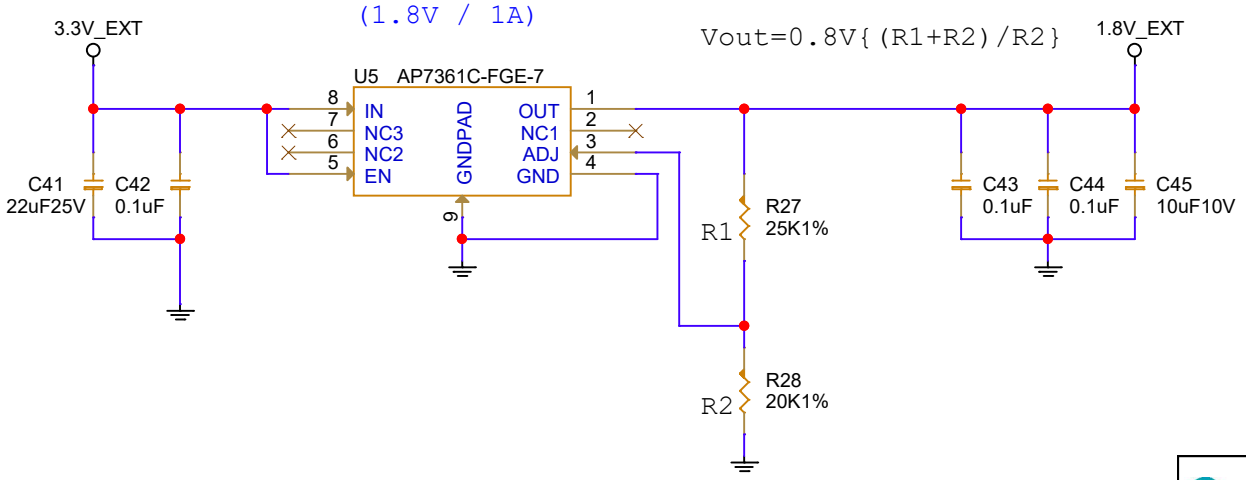
Vout = [(1.43 / Rset) + 1] x 0.6

Vout	Rset	R(RT)	fsw
0.95V	2.45K	1M	250KHz

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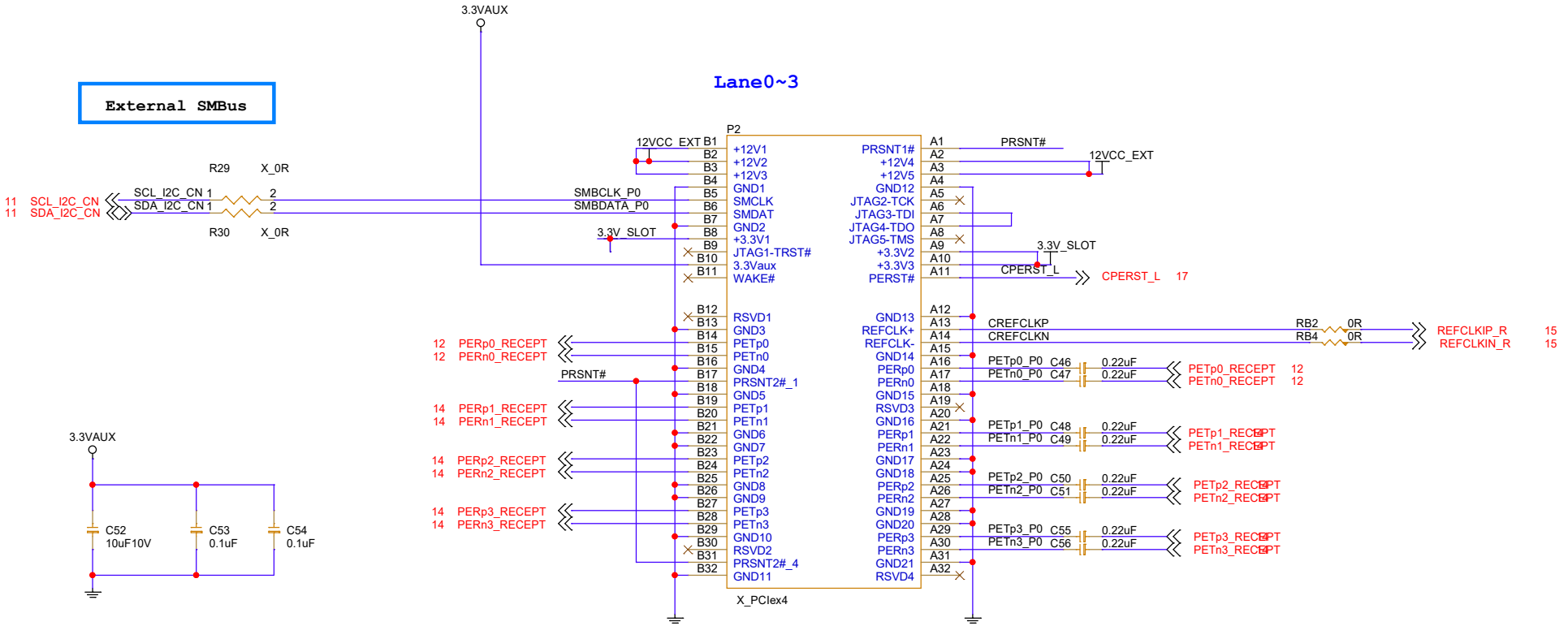
File	PCIe x8 a 4x NVMe M.2 - P0C8X2G808GP	
Size	Document Number	Rev
Custom	06_3.3V_EXT-to-0.95VVDDC_0.95VP	1.6
Date:	Sunday, November 08, 2020	Sheet 8 of 20





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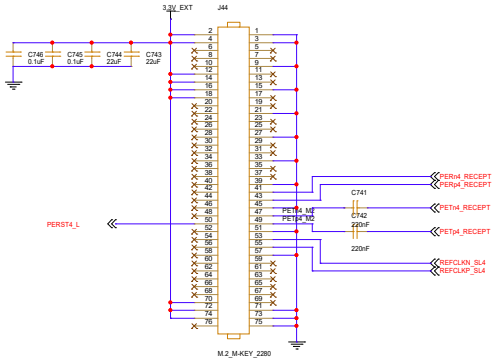
Title: PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP		
Size: A	Document Number: 07_3V3_to_1V8&0.95V_LDO	Rev: 1.6
Date: Sunday, November 08, 2026	Sheet 7 of 20	



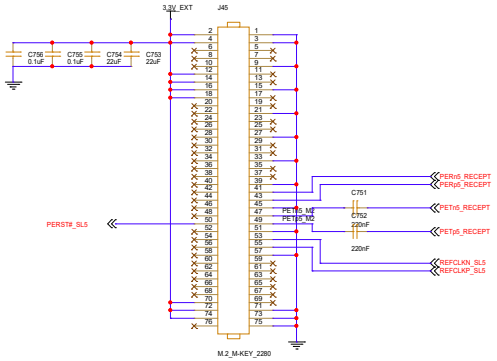
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Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size: Custom	Document Number: 08_UPSTREAM	Rev: 1.6
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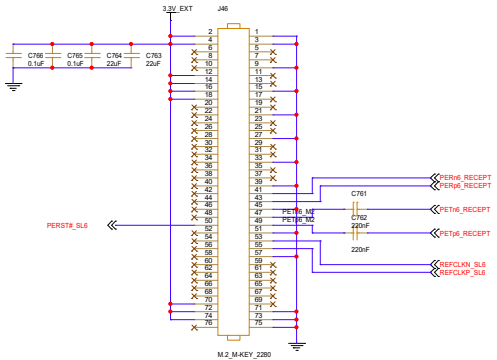
J44 - SSD 1: puerto P4 x1 (lane 4)



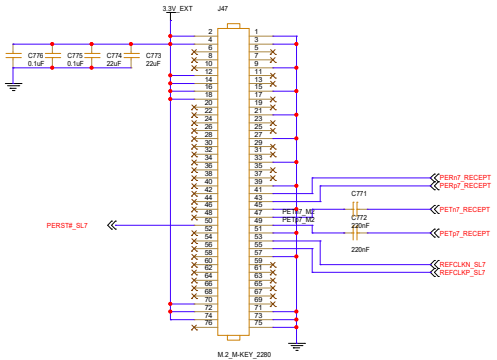
J45 - SSD 2: puerto P5 x1 (lane 5)



J46 - SSD 3: puerto P6 x1 (lane 6)



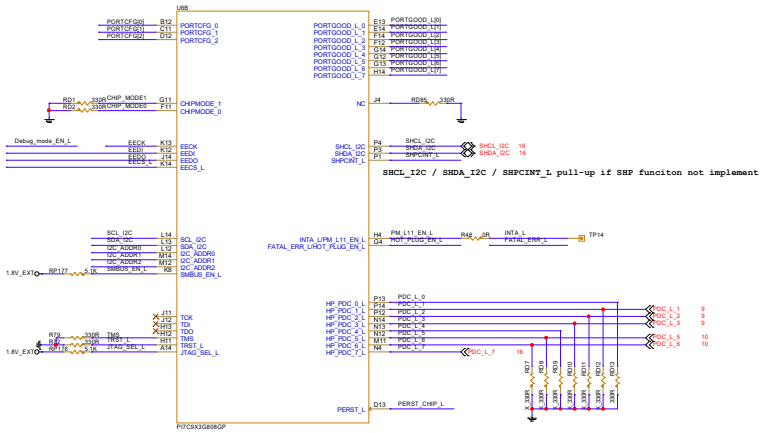
J47 - SSD 4: puerto P7 x1 (lane 7)



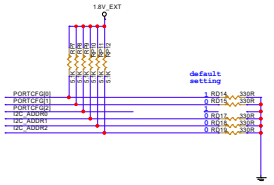
M.2 M-key 2280 (NVMe x4): 3.3V_EXT, caps AC 220 nF en TX, REFCLKO4-7 (55/53), PERST# (50)

Title	PCIe x4 a 4x NVMe M.2 - P17C8X3G809GP	
Size: Custom	Document Number: 10_M.2 SSD 4 - 7	Rev: 1.6
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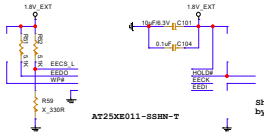
CHIP_MODE[1:0]	Mode
00	Normal
01	Reserved
10	Reserved
11	Reserved



SHCL_I2C / SHDA_I2C / SHPCINT_L pull-up if SHP funciton not implement

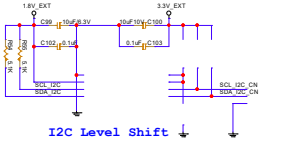


PORTCFG[2]	PORTCFG[1]	PORTCFG[0]	Functional Mode
0	0	0	Reserved
0	0	1	2Port - 8Lane Configuration
0	1	0	3Port - 8Lane Configuration
0	1	1	4Port - 8Lane Configuration
1	0	0	5Port - 8Lane Configuration
1	0	1	8Port - 8Lane Configuration
1	1	0	Reserved
1	1	1	Reserved

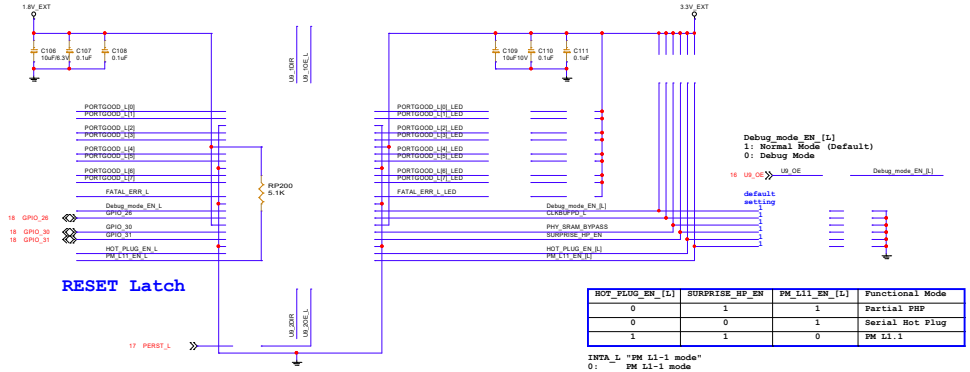


AT25XE011-SSHN-T

Short this connector to bypass the EEPROM



I2C Level Shift



Debug_mode_EN_[L]
1: Normal Mode (Default)
0: Debug Mode

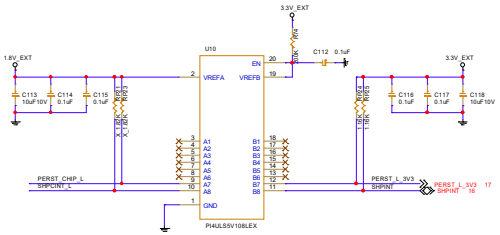
RESET Latch

HOT_PLUG_EN_[L]	SURPRISE_HP_EN	PM_L11_EN_[L]	Functional Mode
0	1	1	Partial PHP
0	0	1	Serial Hot Plug
1	1	0	PM L1.1

INTA_L "PM L1-1 mode"
0: PM L1-1 mode

FATAL_ERR_L "HOT_PLUG_EN_L"
0: GPIO[31:0] Parallel hot-plug mode

Level Shift for PERST / SHP

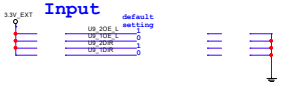


CONTROL INPUTS		OUTPUT CIRCUITS		
OE#	DIR	A PORT	B PORT	OPERATION
L	L	Enable	Hi-Z	B-> A
L	H	Hi-Z	Enable	A-> B
H	X	Hi-Z	Hi-Z	Isolation

GPIO[26] "CKBUFPD_L"
1: Enable CLK buffer
0: Disable CLK buffer

GPIO[30] "PHY_SRAM_BYPASS"
1: Bypass SRAM (Default)
0: Not bypass SRAM

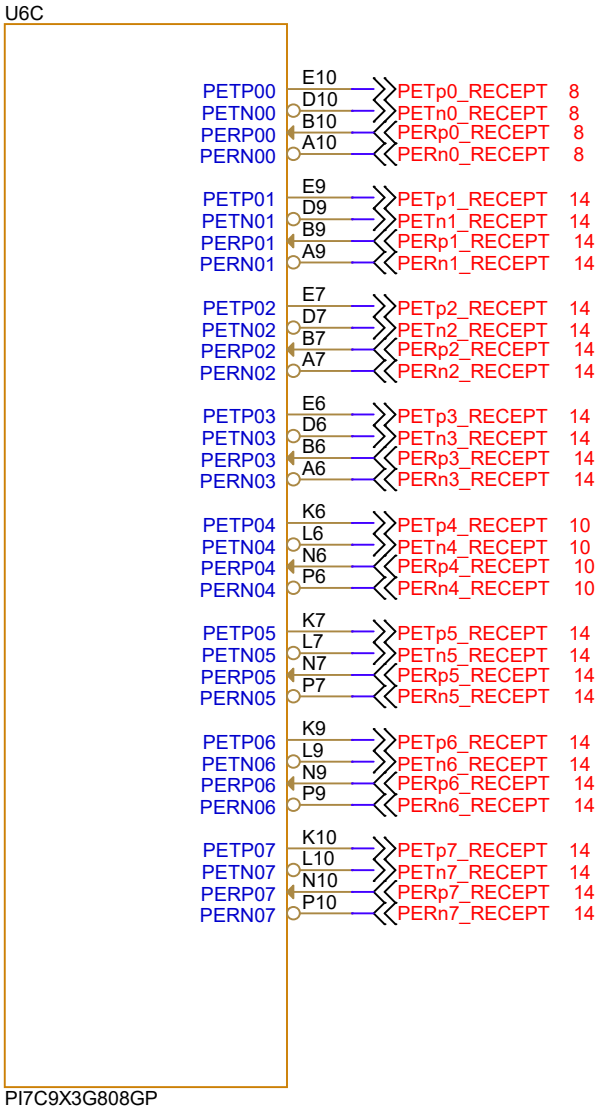
GPIO[31] "SURPRISE_HP_EN"
1: Surprised Hot-Plug: LED Management / Partial PHP (no IO expander or CPLD is found)
0: Managed Hot-Plug: PCIe SHP / PCIe PHP (no IO expander or CPLD is found)

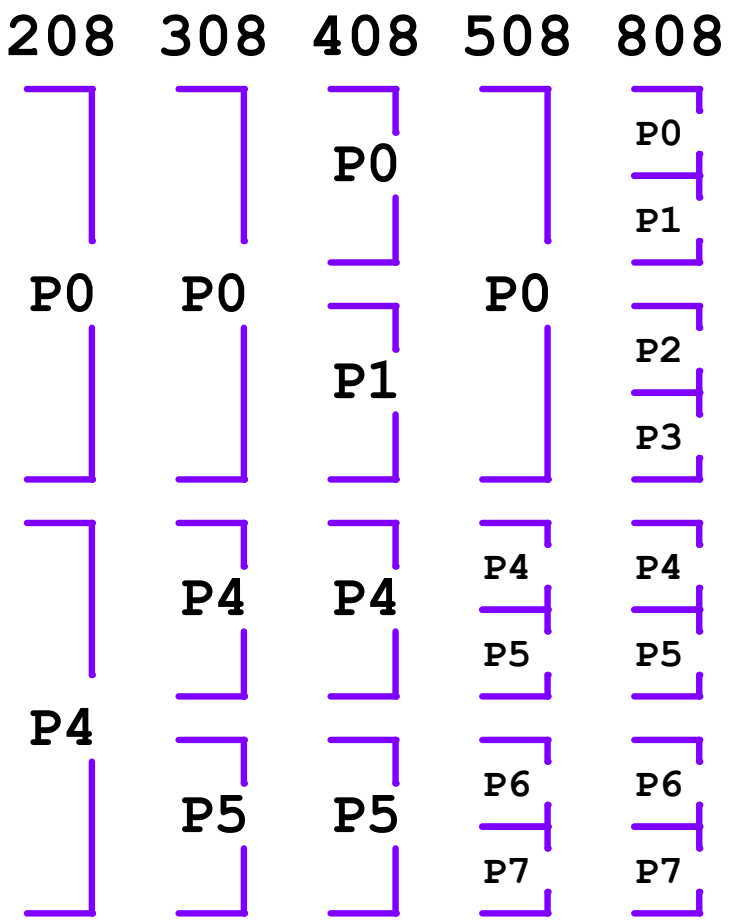


The control pins are supplied by VCCA

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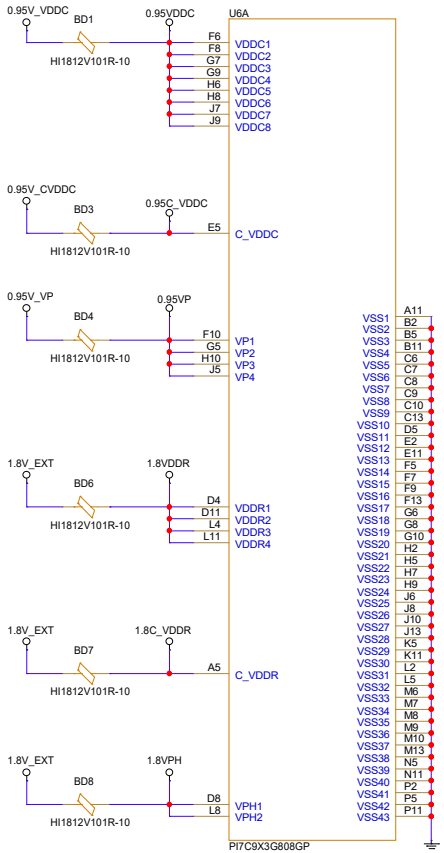
File	PCIe x4 x4c NVMe M.2 - PI7C9X3G808GP		
Size: Custom	Document Number: 11_PI7C9X3G808GP MODE SEL		Rev: 1.0
Date: Sunday, November 08, 2026	Sheet 11 of 20		

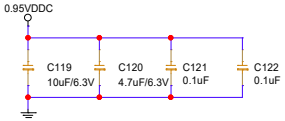


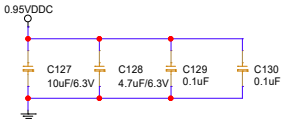


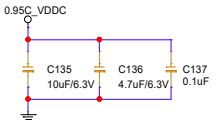
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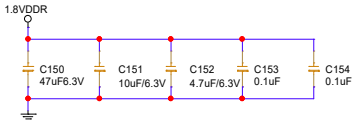
Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP	
Size A	Document Number 12_PI7C9X3G808GP DIFF PAIR	Rev 1.6
Date: Sunday, November 08, 2026	Sheet 12 of 20	

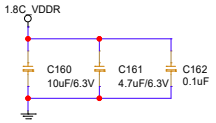


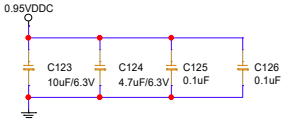


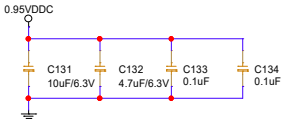


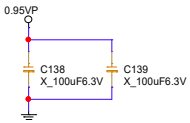


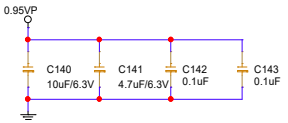


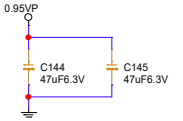


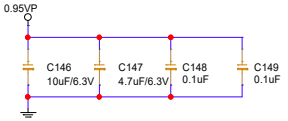


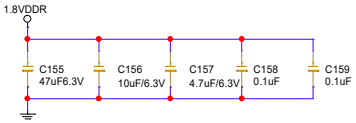


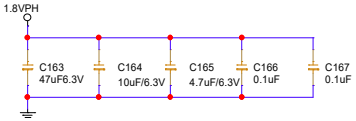




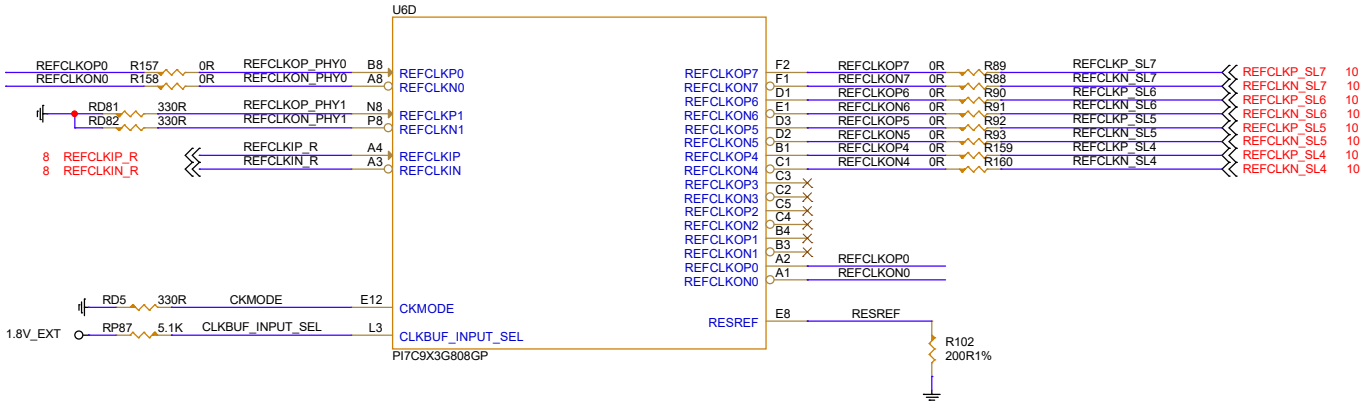




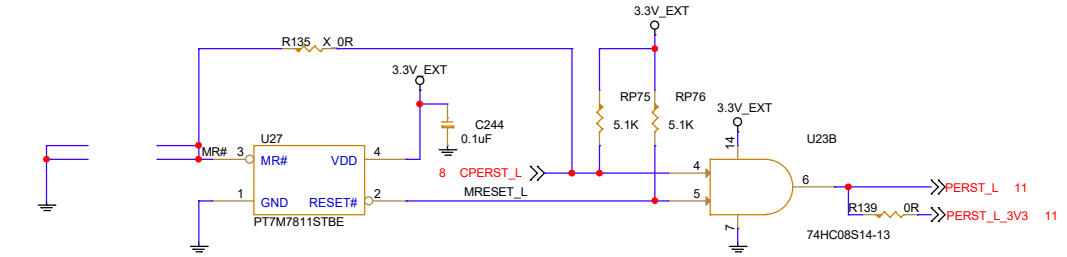


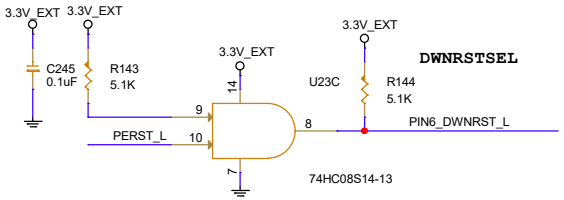


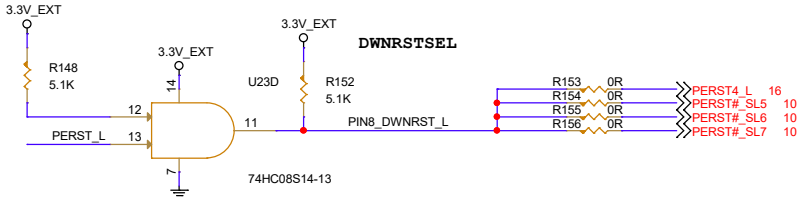
Title	PCIe x4 a 4x NVMe M.2 - PI7C9X3G808GP		
Size: Custom	Document Number: 13_PI7C9X3G808GP POWER		Rev: 1.6
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REVISION HISTORY - PCIe x4 a 4x NVMe M.2

REV	DATE	DESCRIPTION	PAGE	EDITOR
1.0	20_06_02	New design version	All	Eros
1.1	20_06_22	1. U6.J4 pin name change to TEST 2. Add pull low resistor RD85 at U6.J4	10	Eros
1.2	20_07_13	1. 0.9V change to 0.95V 1-1. R4 value change from 2.4K to 2.45K 1-2. R23 value change from 2.4K to 2.45K 1-3 Power source change text from 0.9V to 0.95V	06 13	Eros
1.3	20_09_02	1. C215 & C216 change part to R183 & R184 2. Add note for REFCLKIP/N input	15	Eros
1.4	21_01_20	1. Remove Debug circuit 2. CHIPMODE fix to 00b (Normal Mode) 3. CKMODE fix to 0, REFCLKP/N_1 connect to GND via 330R 4. Remove external clock buffer part. 5. SSC Isolation function / circuit removed	All	Eros
1.5	21_02_09	1. Change pin name from HP_PDC_L_[x] to PDC_L_[x]	9,10 11, 16	Eros
1.6	21_07_07	1. Change CLKBUFPD_L pin default settiing value from 0 to 1	11	Eros
1.0PR	26_11_08	PCIe x4 a 4x NVMe M.2 1. Cold firmware x4 directo a U6 2. Fuente 12V directo a J44-J47 3. U14/U15/U17 3, fuente ATX, 4. EEPROM manual 02-05, portada y title blocks	All	JRQD

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Size: Custom	Document Number: 19_REVISION HISTORY		Rev: 1.6
Date: Sunday, November 08, 2026	Sheet 19 of 20		

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Size: Custom	Document Number: 20_DISCLAIMER	Rev: 1.6
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